Male fiddler crabs: preliminary particle EDA

Scope: two videos, 28,800 frames each, 7,985,830 source detections; nominal sampling 29.970 Hz. Each movie spans 960.96 s including the final frame interval. No females, burrows, physical distance calibration, or validated long-term animal identities are available.

This report separates spatial statistics of accepted male-targeted detections from motion statistics conditional on experimental tracklet continuity. Coordinates are Shell landmarks (explicit centroid proxies) in 7680 x 4320 original-image pixels, not physical body centers. Results describe this model output, not a census of uniquely identified males.

Crab_Cam_2_2: 121.1 accepted land detections/frame on average; nearest-neighbor median 148.9 px. At 1.001 s, displacement median 1.64 px, RMS 7.05 px, alpha2 9.80. At 10.01 s, only 54.6% of sampled land origins retain a usable uninterrupted path.

Crab_Cam_2_3: 141.2 accepted land detections/frame on average; nearest-neighbor median 147.9 px. At 1.001 s, displacement median 2.06 px, RMS 10.59 px, alpha2 5.13. At 10.01 s, only 45.4% of sampled land origins retain a usable uninterrupted path.

Main interpretation: spatial occupancy and short-time displacement distributions are accessible with male data alone. Non-Gaussian displacement statistics, count/density memory and spatial pair structure are useful diagnostics, but neither social forces, territorial organization, nor biological diffusion can be inferred from these summaries alone. Localization jitter, persistent false detections, common image motion and survivor selection remain alternatives.

Reproducibility: CPU Slurm job 15600144; fixed seed 20261015; 16 known-answer unit tests passed before the real-data analysis, plus input and output validation. Source tables, predictions, trackers and models were not modified.

Findings from the reviewed plots

1. Heterogeneous apparent mobility. The measured radial van Hove densities contain a large near-zero component and a much broader tail than the matched-MSD Gaussian benchmark. At 1.001 s, 90.7% / 82.7% of land pairs move <10 px in Cam 2_2 / Cam 2_3. The strong positive alpha2 values (9.80 / 5.13) are inconsistent with that simple Gaussian description of the observed distributions; they identify no specific biological mechanism.

2. Larger displacements in Cam 2_3. Its land MSD is 2.26 times Cam 2_2 at 1.001 s. This difference persists under the fixed-origin and endpoint-score checks. It is a difference between these recorded/model-derived samples, not a population-level comparison. Survival selection is more consequential than the endpoint-score threshold here. Crab_Cam_2_2: restricting to the same origins surviving 10.01 s lowers the 1.001-s MSD by 26.9%; the >=0.8 endpoint-score check changes it by +1.1%. Crab_Cam_2_3: restricting to the same origins surviving 10.01 s lowers the 1.001-s MSD by 17.4%; the >=0.8 endpoint-score check changes it by +1.2%.

3. Spatial concentration largely follows the occupancy footprint. Rectangle-corrected g(r) is elevated across much of 0-2000 px, whereas the land same-time/time-mixed ratio is visually close to one beyond roughly 200 px. At 50-100 px the ratios are 0.61 / 0.70; at 100-200 px they are 0.95 / 0.94. This contrast warns against calling the uniform-null clustering social attraction. The short-distance deficit also cannot establish avoidance because time-mixed self-persistence is unresolved.

4. Persistent dynamics, but nonstationary counts. Connected velocity correlations remain positive: 0.43 / 0.51 at 1.001 s and 0.13 / 0.21 at 10.01 s. Connected coarse-cell density correlations decay over tens of seconds; removing total-count variation changes them little. The large long-lag count correlation in Cam 2_3 is strongly reduced by linear detrending. Count memory is therefore not a clean collective relaxation time.

5. Spacing-speed and resolution warnings. More isolated detections have lower median 1-s apparent speed in the plotted bins, but overlapping IQRs, spatial confounding and encounter-related track loss preclude a causal conclusion. Both van Hove plots show a shared feature around the one-pixel scale; this deserves a localization / prediction-discretization check, not interpretation as a preferred biological step.

Methods and estimands

Selections. Raw counts retain every detection. Spatial and motion primary selections use accepted_for_tracking=True (finite, in-image Shell; body score >=0.5; duplicate-candidate rejection). Land means the accepted fixed-polygon water label is explicitly false, not missing. All accepted detections are retained as a sensitivity comparison. Land removal splits continuous segments; no displacement crosses a gap, water interval, video boundary or tracklet ID change.

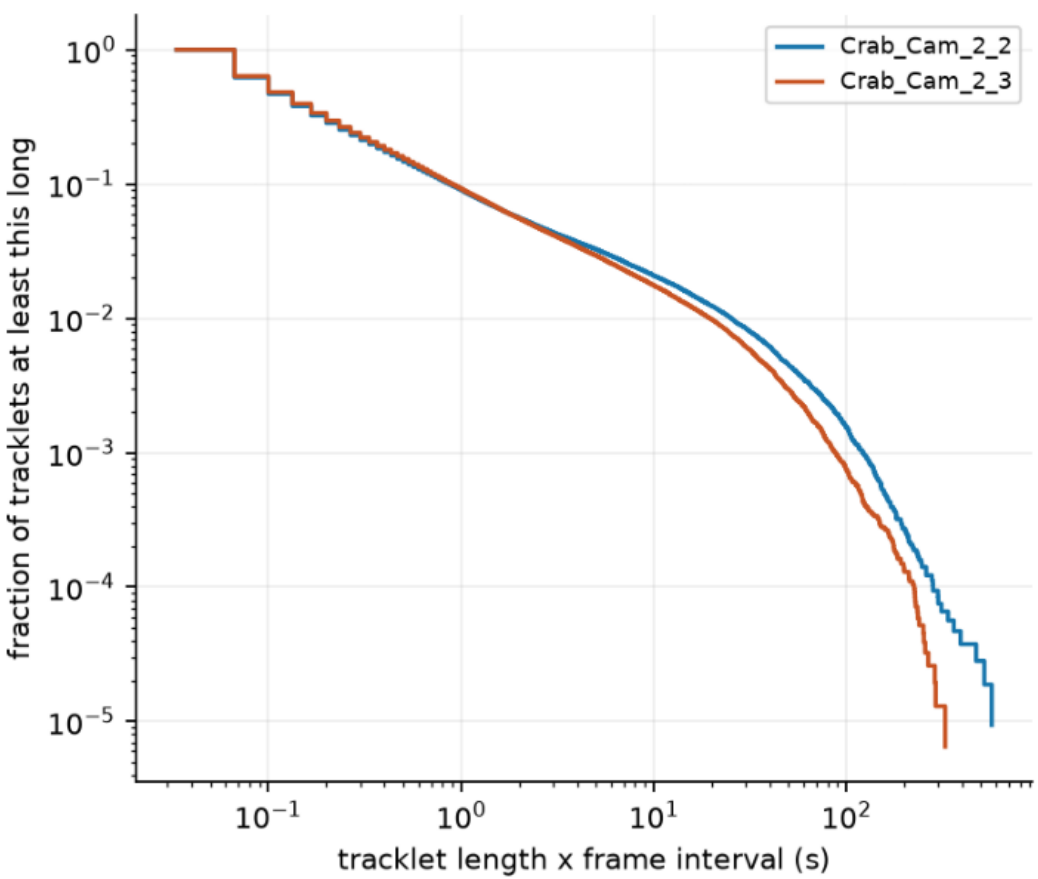
Sampling. Counts use every frame. Spatial statistics use frames 0,30,...,28770 (960 frames/movie; 1.001-s spacing). Motion uses time origins at original frame multiples of 10, with lags 1,3,10,30,60,150,300,600,1500 frames (0.0334 to 50.05 s). Every retained origin-endpoint path must remain continuously observed. All measurements are unsmoothed. A fixed-origin comparison holds the eligible origins at the 300-frame lag constant for all shorter lags.

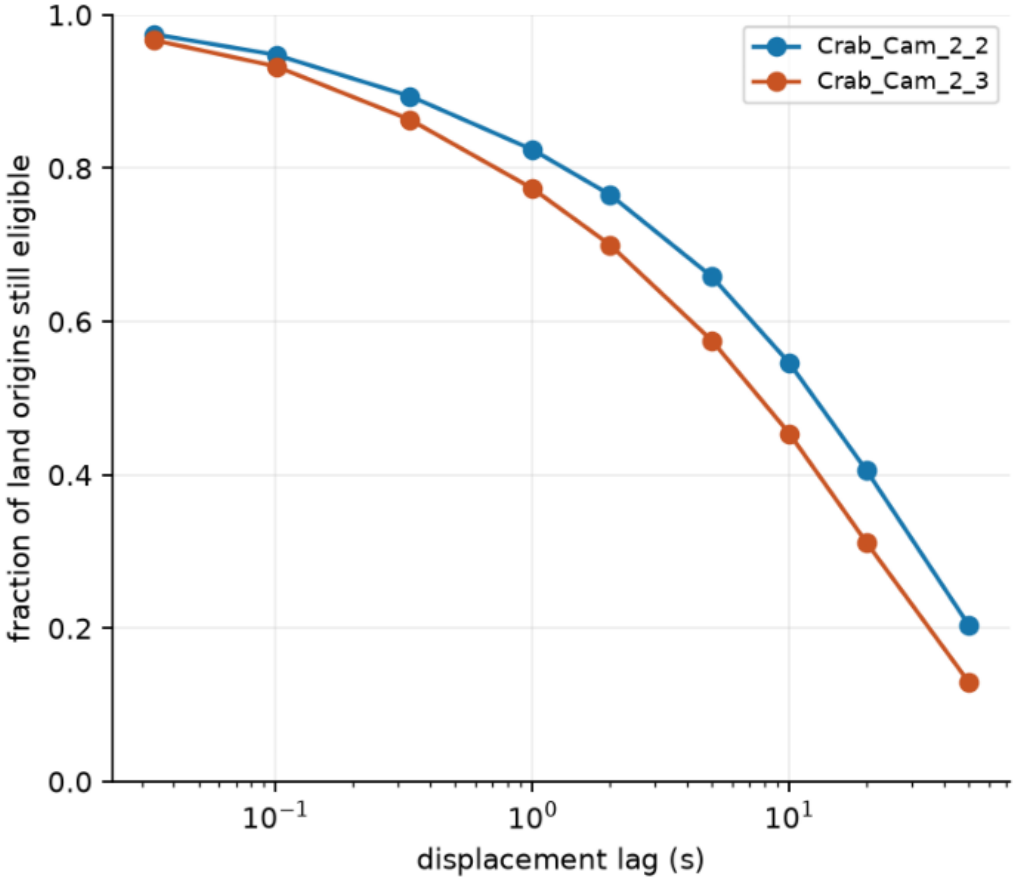
Motion definitions. MSD(tau) = mean |x(t+tau)-x(t)|^2. The self van Hove function is shown as the radial probability density p(r;tau) = 2*pi*r*G_s(r;tau) under angular averaging, not G_s per unit area. Histograms include zero displacement and integrate to one. The isotropic zero-drift Gaussian reference is p(r) = 2r/MSD * exp(-r^2/MSD). alpha2 = mean(r^4)/(2*mean(r^2)^2)-1; alpha2=0 for that Gaussian. Q(10 px,tau)=P(r<10 px). Drift is not subtracted from displacement moments.

Uncertainty. MSD bands are the 2.5th/97.5th percentiles of 300 pooled-ratio resamples of 60-s origin-time blocks (16 full blocks and a short final block). These are within-video stability diagnostics, not independent-animal confidence intervals. Long-range dependence, track selection, systematic model error and the short final block limit interpretation. Other plots are descriptive; no p-values, multiple-testing claims, diffusion fits or population-level inference are made.

Velocity. Coarse velocity is [x(t+10)-x(t)]/(10*dt). Reported correlations use centered velocity vectors and an RMS-product denominator on the SAME surviving origin pairs for each lag. Intervals do not overlap at positive tested lags (>=10 frames). This reduces, but does not eliminate, localization-noise sensitivity. Raw correlations are also saved.

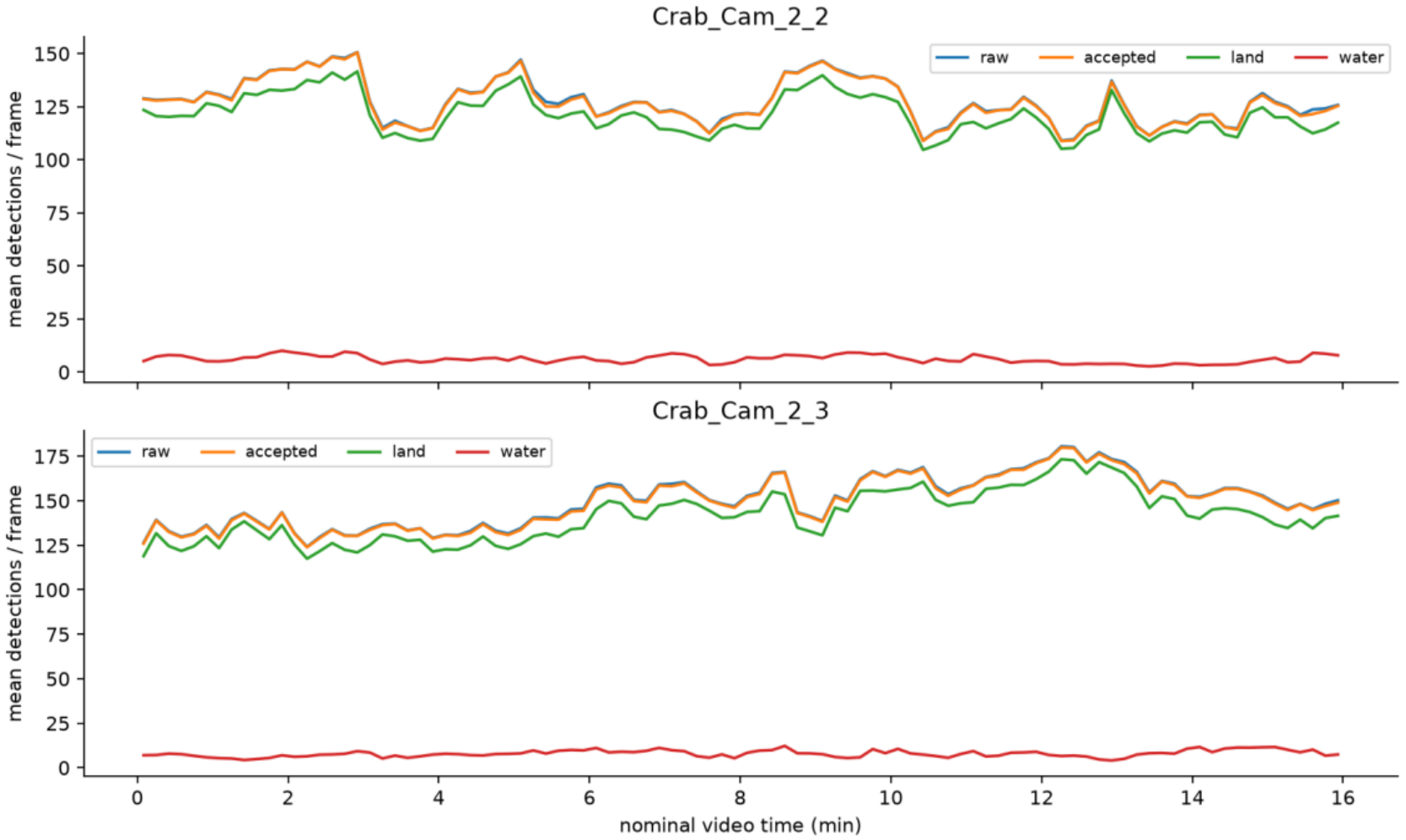
1. Identity support is the first limitation





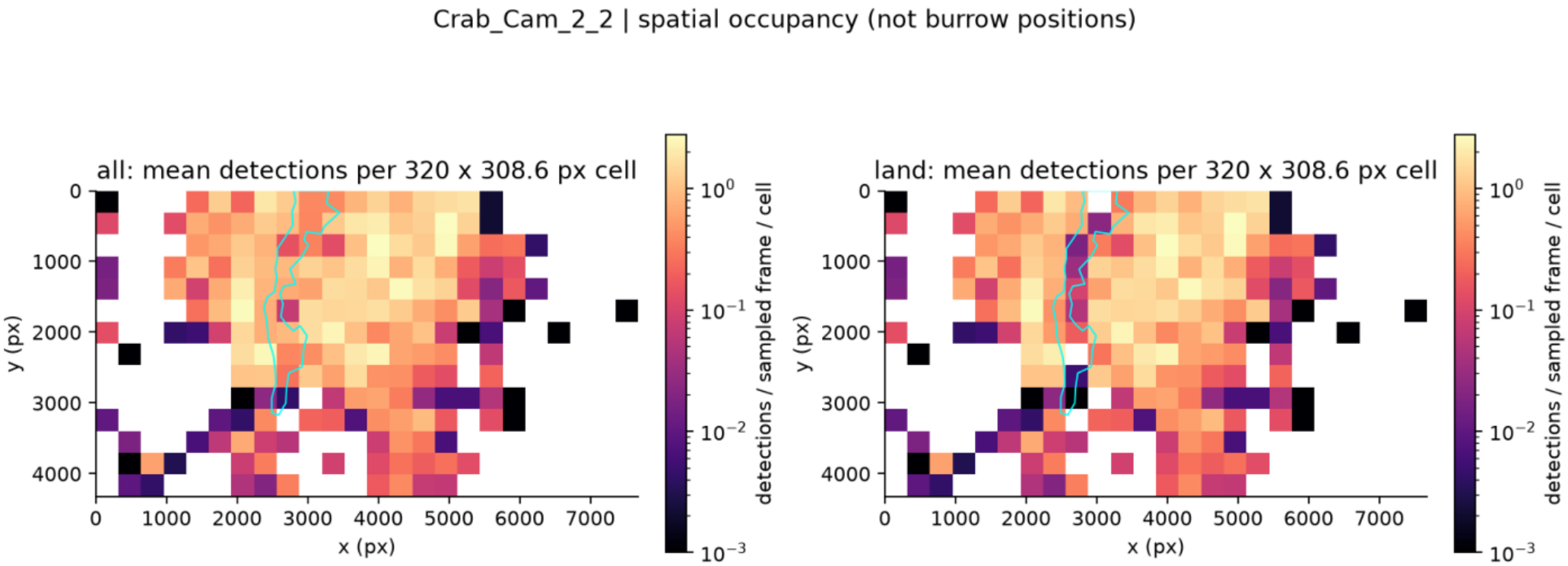
Tracklet survival is not animal lifespan. Most tracklets are short even though long tracks contain many observations. Right: eligible land origin-endpoint pairs divided by all sampled land origins, including origins too close to the end of a video; thus both fragmentation and recording-end censoring reduce support. Long-lag motion is strongly conditional on remaining detected and associated. The tracker forbids single-frame links >12 px, uses velocity prediction, and rejects ambiguous neighbors, so motion and close encounters are selected.

2. Detection counts and nonstationarity



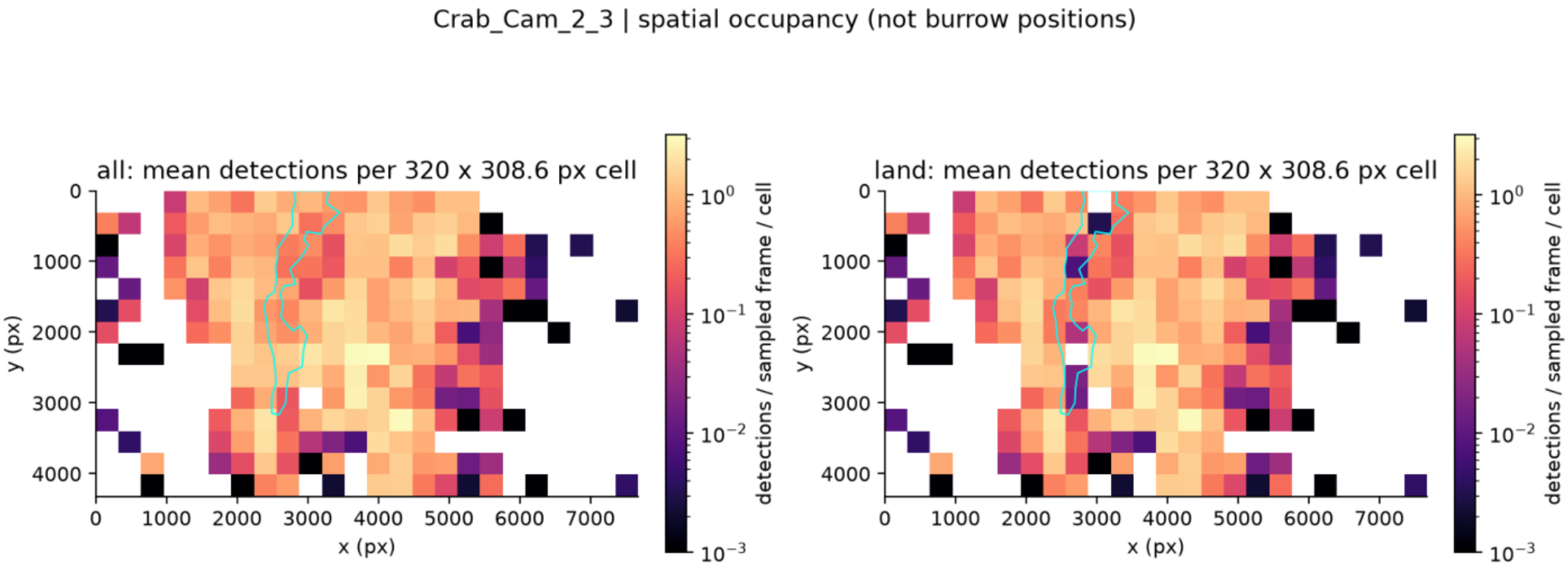
Ten-second display averages; raw calculations use every original frame, including zeros. Raw, accepted and land curves may overlap because relatively few rows are unassigned or water-labeled. Counts are detections, not unique males or population estimates. Crab_Cam_2_2: land first/last nominal minute 122.0/116.7 detections per frame. Crab_Cam_2_3: land first/last nominal minute 125.3/137.9 detections per frame. Changing detection conditions or activity can alter these series without abundance changes.

3. Spatial occupancy: Crab_Cam_2_2



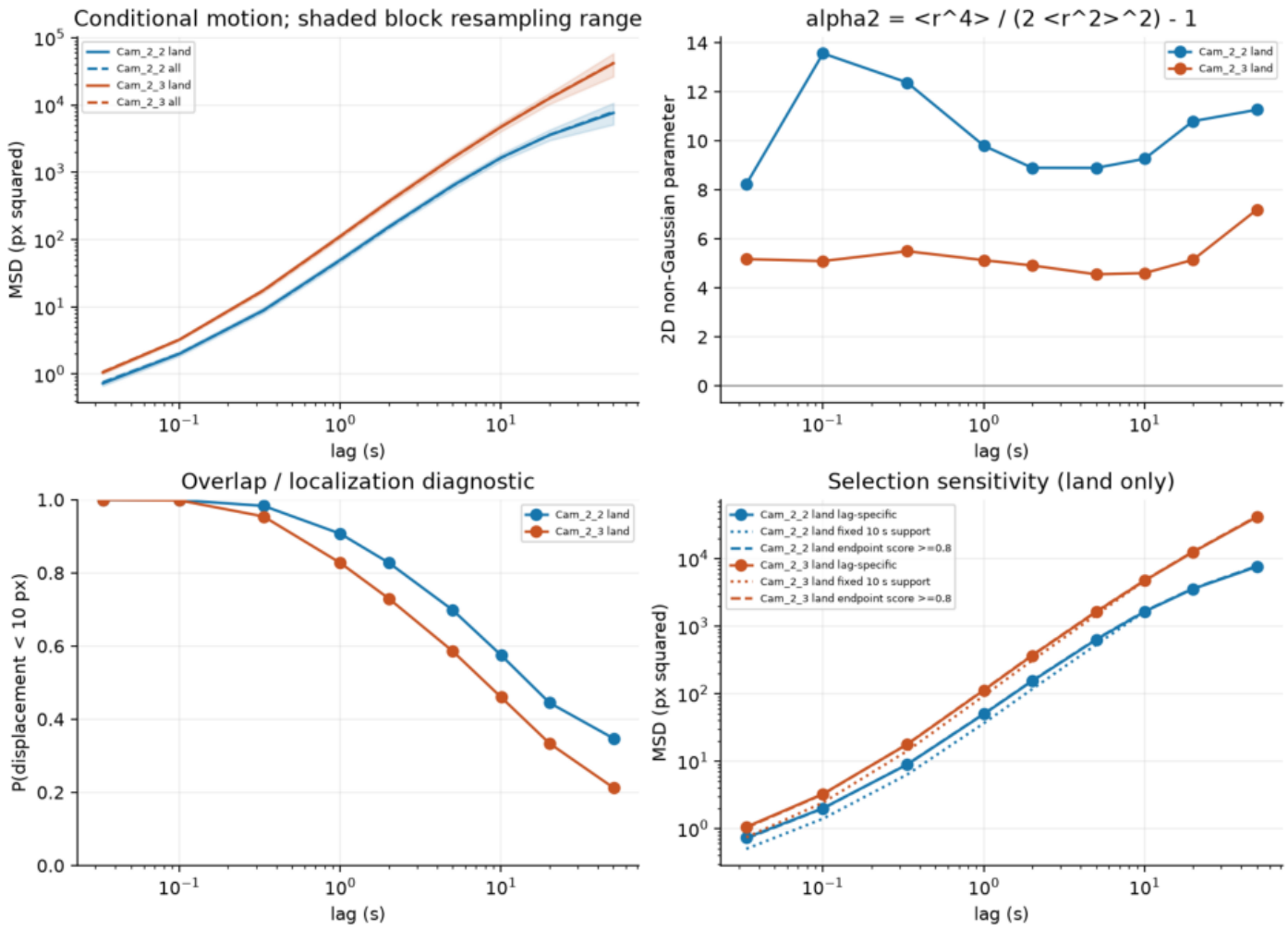
Time-averaged counts on a fixed 24 x 14 grid; each cell is 320 x 308.6 image pixels. Cyan outlines the accepted static water polygon. Color is log-scaled detections per sampled frame per cell, NOT density per square meter or per unit usable habitat. Zero cells are masked. Concentrations can reflect habitat, detection performance or persistent artifacts. Similar spatial maps in the videos do not establish different animals or independent biological replicates. No hotspot is identified as a burrow.

3. Spatial occupancy: Crab_Cam_2_3



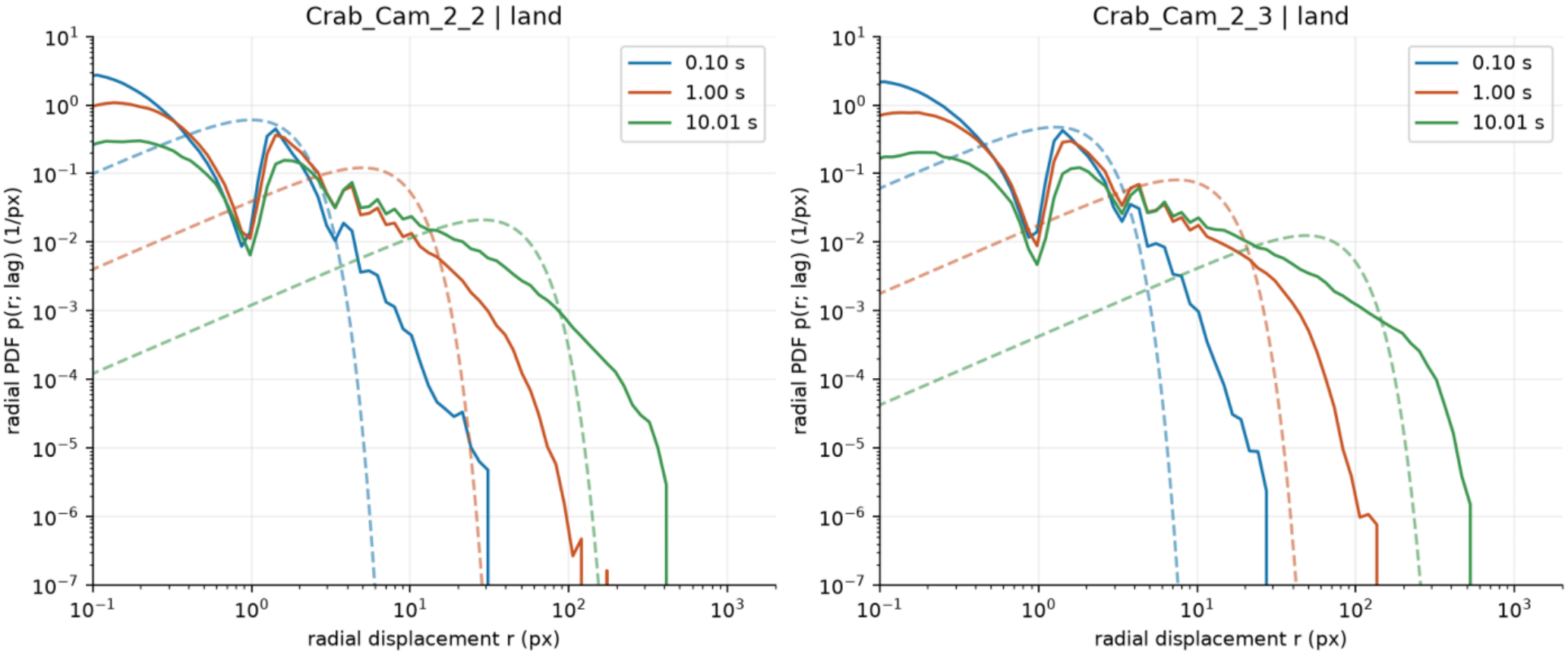
Time-averaged counts on a fixed 24 x 14 grid; each cell is 320 x 308.6 image pixels. Cyan outlines the accepted static water polygon. Color is log-scaled detections per sampled frame per cell, NOT density per square meter or per unit usable habitat. Zero cells are masked. Concentrations can reflect habitat, detection performance or persistent artifacts. Similar spatial maps in the videos do not establish different animals or independent biological replicates. No hotspot is identified as a burrow.

4. Displacement scaling and selection sensitivity



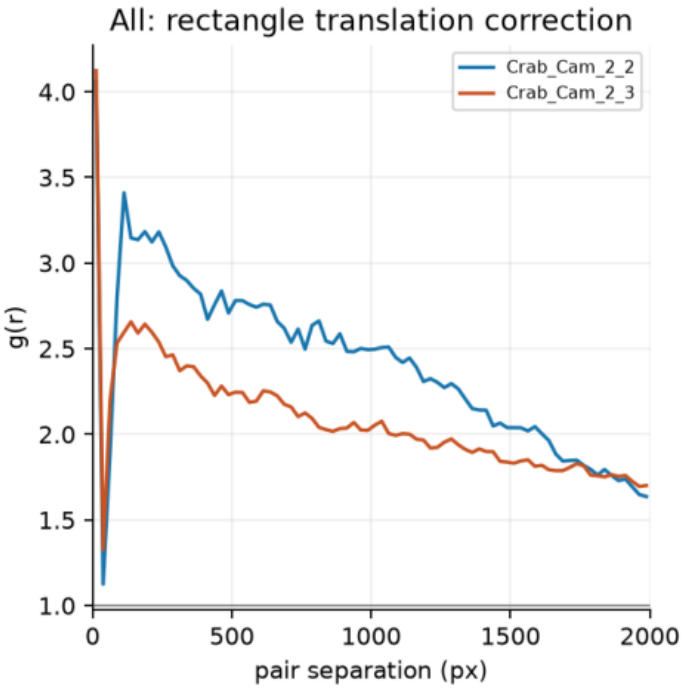
MSD is conditional on uninterrupted experimental identity and is not fitted to a diffusion law. Solid/dashed comparisons separate land and all accepted detections. Alpha2 and the 10-pixel overlap quantify broad displacement distributions and localization. Bottom-right holds origins fixed to paths surviving 10.01 s (dotted) and separately checks >=0.8 endpoint scores (dashed; not a full-path confidence filter). Agreement would not establish identity accuracy; disagreement diagnoses selection. Shading is a temporal-block resampling range.

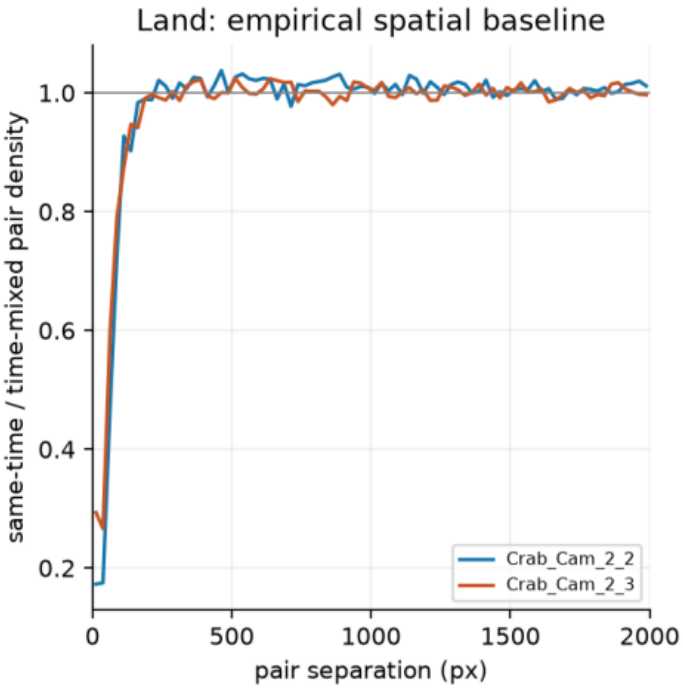
5. Self van Hove radial displacement distributions

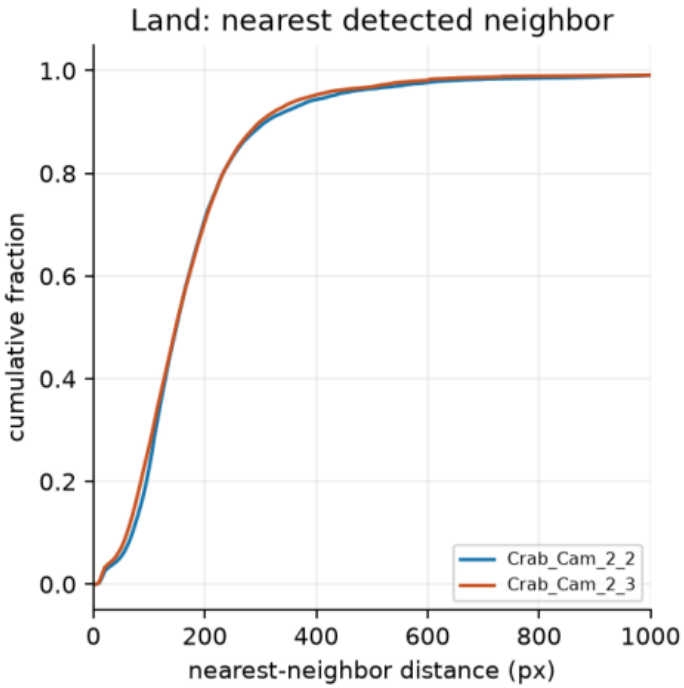


Solid curves are radial displacement probability densities for land segments; dashed curves are isotropic zero-drift Gaussian/Rayleigh references with the same measured MSD. Normalization includes the zero-to-0.1-pixel bin even though the displayed log axis starts at 0.1 px. The plot is p(r), not the area-normalized G_s. Differences can arise from mixed mobilities, pausing, drift, localization error and tracking selection. Heavy tails do not by themselves demonstrate anomalous diffusion or behavioral switching.

6. Pair-distance structure and nearest neighbors







Left: full-rectangle translation-corrected g(r), whose uniform independent null is one; the actual habitat is not uniform. Middle: same-time land pair-distance probability divided by a time-mixed baseline from offsets of 60,120,240 sampled frames (circular), each normalized by its TOTAL possible pairs, not only plotted radii. Right: nearest-neighbor CDF without boundary correction (descriptive only). Crab_Cam_2_2: median nearest land detection 148.9 px; 22.6% have a neighbor <100 px. Crab_Cam_2_3: median nearest land detection 147.9 px; 26.8% have a neighbor <100 px.

Interpreting spatial correlation without burrows

The full-image pair estimate uses unordered distinct pairs and weights each separation (dx,dy) by W*H/[(W-|dx|)*(H-|dy|)]. Weighted annular counts are divided by the sum of N(N-1)/2 over sampled frames, times annular area/(W*H). This corrects rectangular observation-window truncation, not inaccessible habitat, perspective or spatially varying detectability. A uniform-rectangle control verifies the normalization.

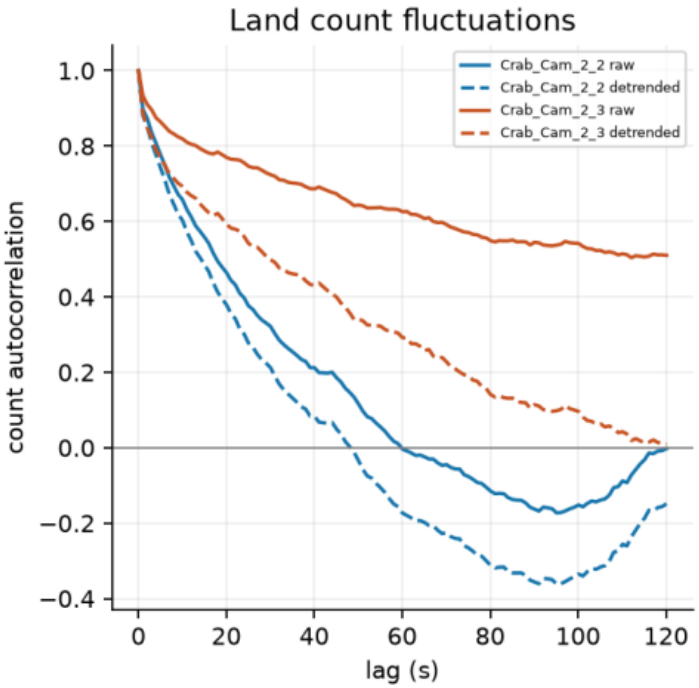
The time-mixed land baseline compares positions in different sampled frames, using cross pairs normalized by sum N(t)*N(t+offset). It preserves the empirical footprint rather than assuming that the entire land area is accessible. Offsets are approximately 60.06, 120.12 and 240.24 s with circular wrapping. Normalization is pair-weighted across frames. It is an exploratory diagnostic, not an inhomogeneous-point-process estimate or a hypothesis-test null.

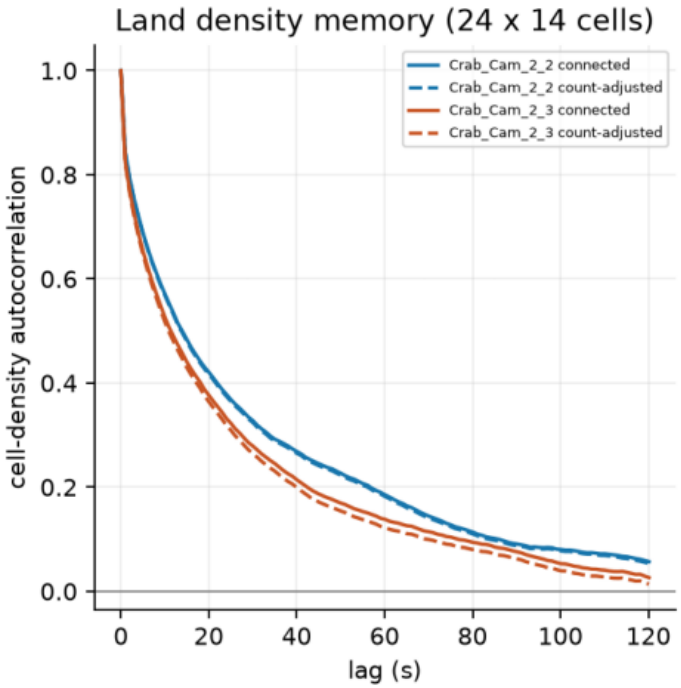
Crucially, time-mixed pairs can contain the SAME stationary male, repeated detections at a burrow, or a persistent artifact at both times. Those cannot be reliably removed using fragmented IDs. A short-distance deficit against this baseline could therefore be self-persistence rather than interpersonal avoidance. Nonstationarity also changes the baseline. The rectangle and time-mixed plots should be read together.

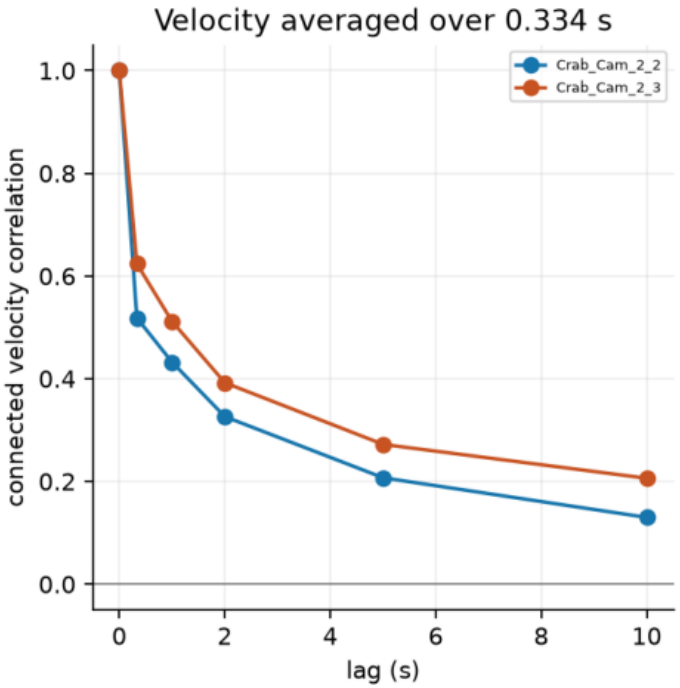
At very short distances, the tracking acceptance rule rejects bodies within 5 pixels as duplicate candidates. Missed close individuals, unresolved body overlap and tracker ambiguity add selection. Nearest-neighbor distances near image borders may be overestimated because neighbors outside the image are unseen. Neither a preferred spacing nor a repulsion scale can yet be assigned a biological cause.

The most useful next spatial additions are a usable-habitat mask, physical/perspective calibration, explicit burrow coordinates and representative detection audits. Female positions would enable male-female cross-correlations; they are not necessary for these descriptive male-only statistics, but are necessary for courtship-specific claims.

7. Temporal count, density and velocity memory

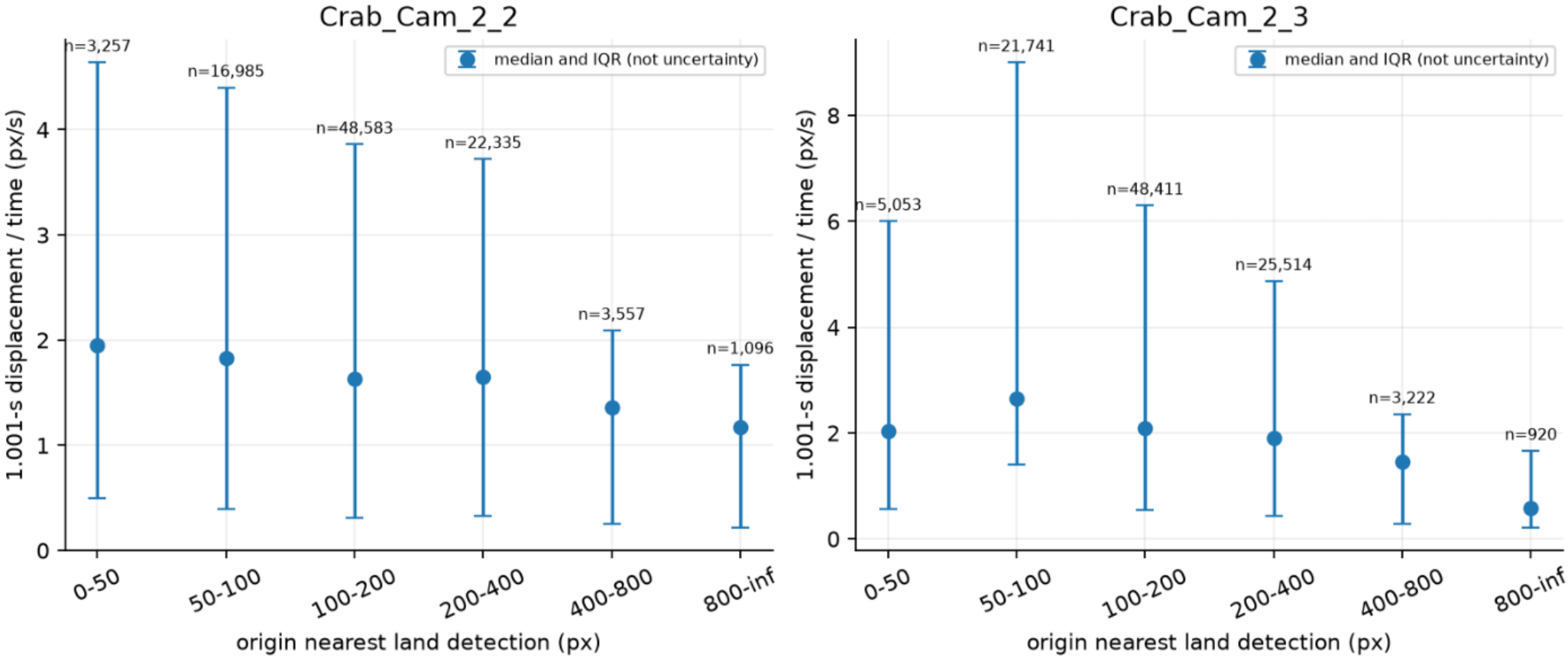






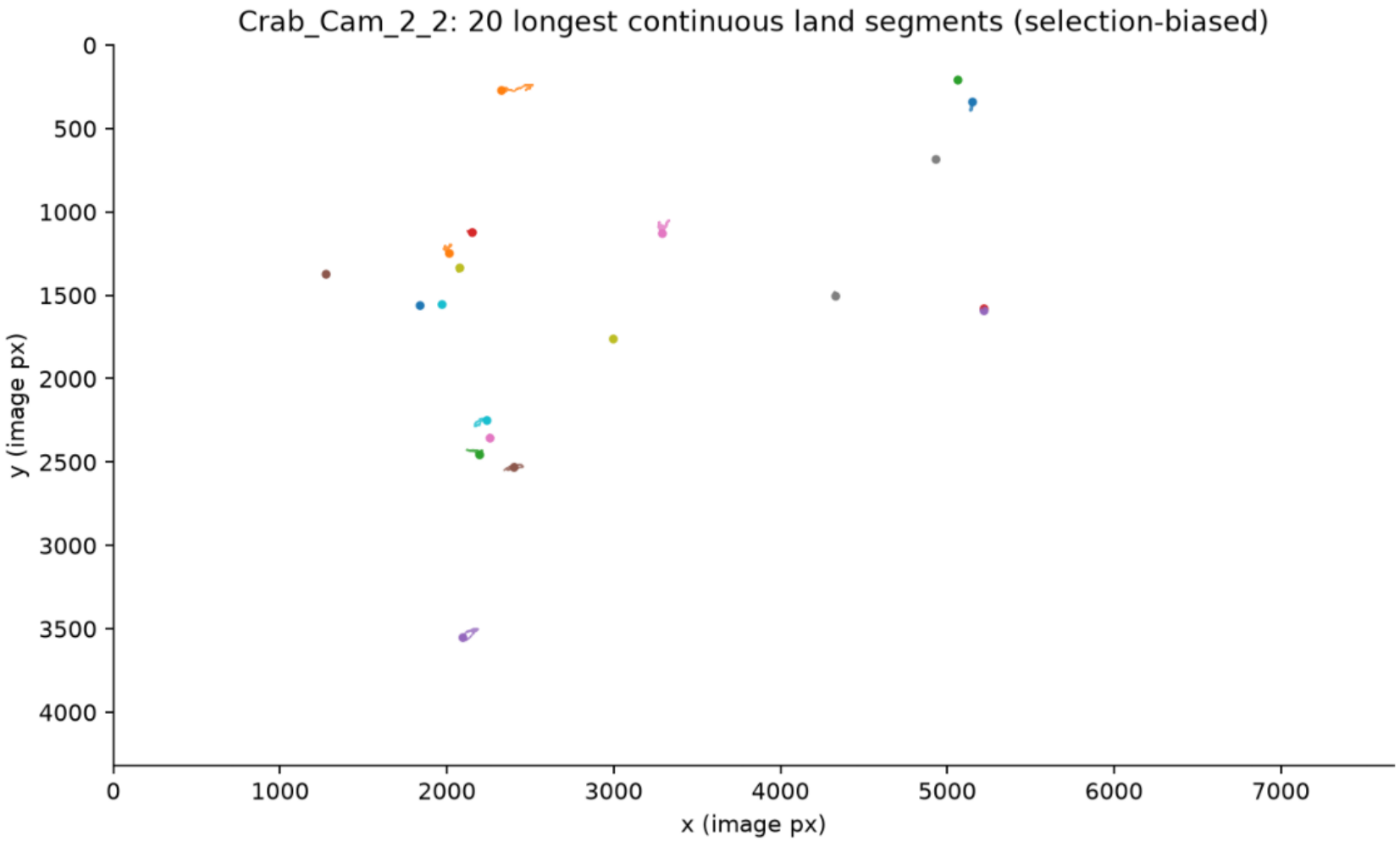
Count ACF subtracts the movie mean; dashed also removes a linear time trend. Cell ACF subtracts each cell mean and sums covariance over 24 x 14 cells. The count-adjusted version subtracts N(t) times the mean occupancy profile before centering, separating spatial redistribution from total-count changes. ACF uses available pairs at each lag and lag-zero variance. Velocity correlations use continuously tracked land segments. Positive memory need not be social coordination; persistent occupancy and detection conditions also generate it. Adjacent velocity increments share endpoint localization noise.

8. Local spacing versus apparent speed



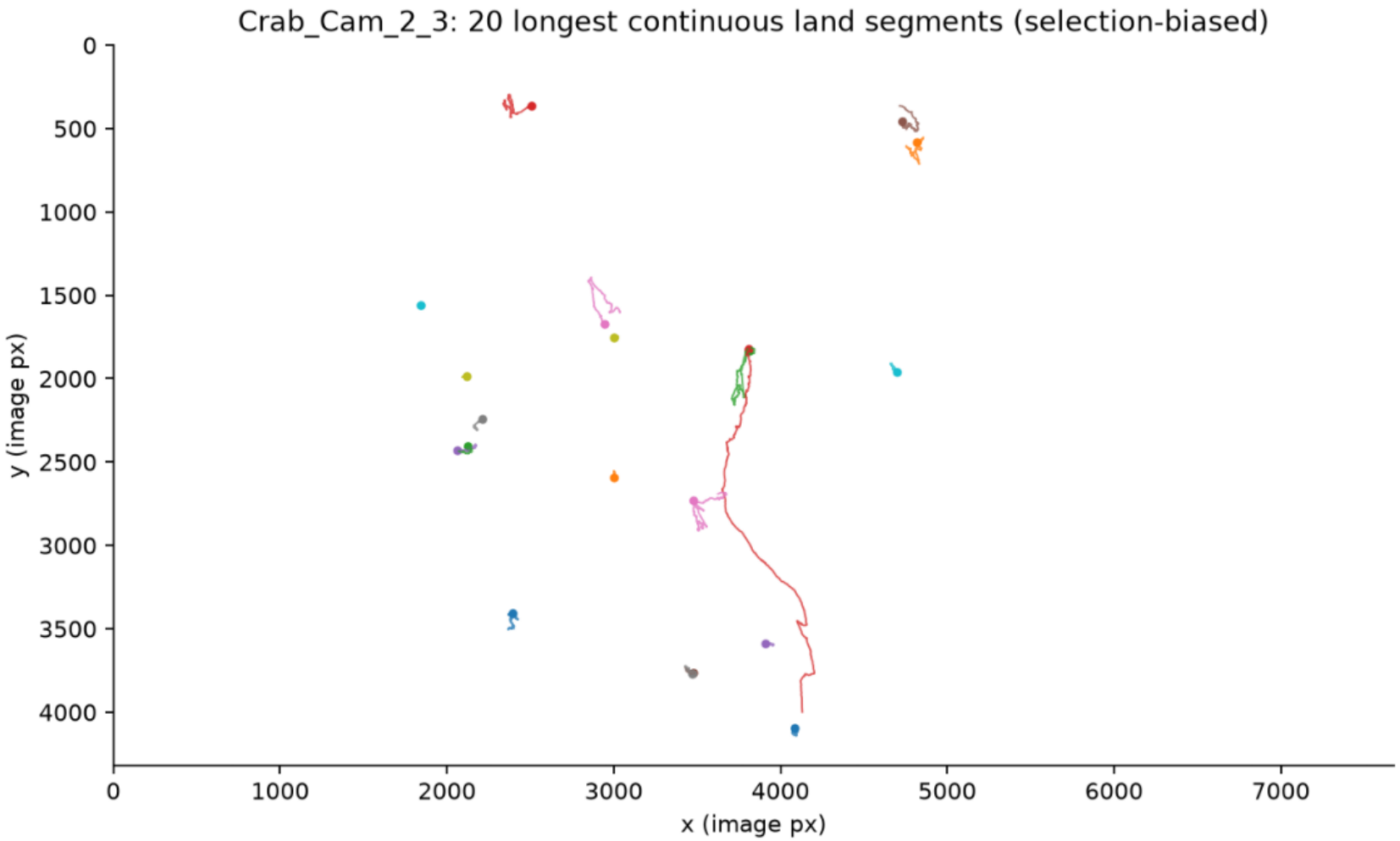
For accepted land detections at sampled frames, the closest OTHER accepted land detection is found at the origin; speed is the 30-frame displacement / 1.001 s on a surviving land segment. Points show medians, bars the interquartile range, not confidence intervals; n counts correlated origin pairs, not animals. Bins are fixed in image pixels and the final bin is unbounded. Any trend is descriptive and may reflect habitat, perspective, encounter-related fragmentation or misdetections, not a causal effect of density on behavior.

9. Trajectory examples: Crab_Cam_2_2



Twenty longest continuous land segments, deliberately selected for visibility; points mark their starts. Displayed vertices are sampled every 10 frames, without smoothing or gap bridging in the analysis. Axes are absolute image coordinates with y increasing downward. These examples are not representative random animals and have not been validated against video ground truth. Strong localization may reflect site fidelity, immobility, limited duration or a persistent false detection.

9. Trajectory examples: Crab_Cam_2_3



Twenty longest continuous land segments, deliberately selected for visibility; points mark their starts. Displayed vertices are sampled every 10 frames, without smoothing or gap bridging in the analysis. Axes are absolute image coordinates with y increasing downward. These examples are not representative random animals and have not been validated against video ground truth. Strong localization may reflect site fidelity, immobility, limited duration or a persistent false detection.

Quantitative synopsis and limits

Crab_Cam_2_2: at 0.1001 / 1.001 / 10.01 s, land MSD = 2.00 / 49.73 / 1646.86 px^2; alpha2 = 13.57 / 9.80 / 9.27. At 1.001 s, 90.7% of retained displacement pairs are <10 px (287,483 origin pairs; 7,869 contributing segments). The all-accepted one-frame step maximum is 11.9927 px.

Crab_Cam_2_3: at 0.1001 / 1.001 / 10.01 s, land MSD = 3.25 / 112.25 / 4729.68 px^2; alpha2 = 5.10 / 5.13 / 4.61. At 1.001 s, 82.7% of retained displacement pairs are <10 px (314,689 origin pairs; 11,626 contributing segments). The all-accepted one-frame step maximum is 11.9806 px.

The Gaussian reference is a benchmark, not a fitted mechanistic model. Nonzero drift also changes alpha2 relative to a centered isotropic Gaussian. Pixel localization noise contributes an MSD plateau: independent per-coordinate error variance sigma^2 produces MSD=4*sigma^2 even for a stationary body. Motion near this scale cannot be interpreted as locomotion without a localization-error calibration.

Longer paths select easier-to-track, often more stationary detections. The tracker also imposes a 12-pixel single-frame step ceiling (~359.6 px/s), a predicted-position gate, matching costs and ambiguity checks. Thus the observed high-speed tail and close-encounter dynamics are not unbiased. Endpoint score sensitivity is not calibration: model scores are not probabilities of a correct detection.

No female interaction, burrow attachment, mating behavior, true animal abundance, diffusion coefficient, collective phase or social attraction/repulsion is claimed. The two files are analyzed separately and are not assumed to be independent fields or independent male populations. No cross-video identity linkage is attempted.

Tests, validation and reproducibility

All 16 unit tests passed before analysis. Known-answer controls cover stationary and constant-velocity trajectories; Gaussian diffusion (MSD=8*lag for component step variance 4); localization-noise plateau; gap and identity isolation; water-filter splitting; fixed-origin support; a PDF atom at zero and Rayleigh CDF; exact three-point pair/nearest-neighbor geometry; uniform-rectangle g(r) and independent time-mixed ratio; white-noise and AR(1) ACFs; cell mean invariance; constant-value block resampling; rotational invariance; invalid data.

Real-data checks: completion marker and manifest checksum; full SHA-256 comparison of every consumed observation part, frame and tracklet table, and water geometry against the published manifest; exact frame-level raw/assigned count reconciliation; assigned coordinate/ID validity; consecutive original frames within IDs; land/water count accounting; global row accounting; no gap pairs; all one-frame steps <=12 px; all displacement histogram counts retained. Checks fail the job rather than silently filling gaps, clipping positions, changing tracking or dropping bad outputs.

Software checks establish estimator correctness and internal consistency, NOT detection accuracy or true identity. Controls are synthetic and do not capture structured field-video errors. No random identity review or calibration set was available for this analysis.

Run directory: analyses/male_particle_eda_v001/run_15600144/. summary.json gives headline quantities; tables/ contains numerical curves and density grids; figures/ contains PNG/PDF plots; report_pages/ contains page previews; tests.json records the validation receipt; provenance.json pins inputs, code and package versions. COMPLETE.json lists output checksums after successful report creation.

Dataset: datasets/v007_best_body_first_dataframe_v001/. Canonical schema/method references: docs/datasets/v007_best_body_first_dataframe_v001/README.md and docs/BODY_FIRST_TRACKLETS.md. Analysis source and finite Slurm entry point are analysis.py, report.py, test_analysis.py and run.sh in the analysis directory. This is a preliminary, reproducible diagnostic report rather than a validated biological inference.